

MUHAMMAD TAHA BIN ZAEEM

Computer Engineering Undergraduate | NUST CEME

Research Engineering | Trustworthy AI | Computer Architecture | Accessible Computing

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Updated 13 August 2026 | Pakistan

RESEARCH PROFILE

Computer Engineering undergraduate building auditable systems across AI-assisted engineering, computer architecture, and accessible learning. Current independent work develops controlled, reproducible protocols for measuring LLM-induced style change and semantic mutations in EDA project files. Converts open questions into bounded datasets, deterministic pipelines, testable artifacts, and explicit evidence trails.

AREAS OF EXPERTISE

Reproducible research, trustworthy AI, EDA automation, computer architecture, accessible computing

ACADEMIC BACKGROUND

National University of Sciences and Technology (NUST), CEME

Bachelor of Engineering in Computer Engineering | Aug 2025 - Oct 2029 (expected) | Third semester

Selected focus: computer architecture, digital logic, algorithms, programming, circuits, and electronics.

Government College University (GCU), Lahore

Intermediate in Engineering | Grade A | Sep 2023 - Sep 2025

Qazi Grammar Boys High School

Matriculation in Computer Science | Grade A+ | Top Performer | Apr 2021 - Apr 2023

RESEARCH EXPERIENCE

Authorial Style Separability Loss in LLM-Rewritten Fiction

Independent research project | 2026 - Present | Study and manuscript in progress; not yet published

- Defined a controlled stylistic study across lexical, syntactic, rhythmic, and distributional dimensions using six public-domain authors and 12 canonical Project Gutenberg works.
- Built a traceable pipeline covering provenance registries, checksums, conservative cleaning, sentence-aware extraction, and balanced selection of 360 passages from 1,879 candidates.
- Locked paraphrase, modernize, and simplify protocols, output schemas, and quality-control rules before preparing 1,080 rewrite requests; generation and analysis remain ongoing.

Controlled-Mutation Fingerprinting for Semantic Change Attribution in EDA Project Files

Independent research project | 2026 - Present | Phase 1 protocol in development; not yet published

- Designed a ground-truth-first Proteus study using save-only controls, isolated resistor mutations, and topology variants to separate file-save noise from semantic engineering edits.
- Structured a reproducible repository for extraction, file comparison, feature generation, mutation attribution, metadata, and manuscript evidence before analysis begins.

Type2Learn

Founder and Development Lead | Jul 2026 - Present

- Led development of a bilingual, accessibility-first active-learning prototype with learner-controlled layout, colour, motion, sound, read-aloud, and typed or spoken response paths.
- Translated a documented record of 13 interview scripts from nine interviewees plus structured product reviews into feature decisions while preserving dissenting preferences, consent, privacy, and non-diagnostic boundaries.

ENGINEERING AND PROFESSIONAL EXPERIENCE

ProGenEDA

Founder, Product and Engineering Lead | Jul 2026 - Present

- Designed an adapter architecture that converts bounded circuit intent through a canonical CircuitIR into editable native Proteus, KiCad, EasyEDA Pro, and LTspice projects.
- Built evidence-gated validation around 1,000 verified example circuits, with backend-specific checks and explicit release boundaries for structure, connectivity, geometry, native parsing, and simulation claims.

Custom 20-bit Single-Cycle Vector CPU

NUST academic team project | Verilog and Xilinx ISE | 2026

- Co-designed a Harvard-architecture processor with a custom 20-bit ISA, 16-bit scalar execution, and 64-bit SIMD operations across four 16-bit lanes; verified behavior through testbenches and waveforms.
- Owned branch-condition, data-memory, immediate-extension, lane load/store, core integration, and top-level logic; project earned Second Runner-Up in the SEMPEC Junior Hardware category.

PhishRod

Development Intern | Jul 2026 - Present

- Support software development in a security-focused product environment.

Arch Technologies

Software Programmer Intern | Jan 2026 - Apr 2026

- Built four C++17 GUI applications for both Windows and Linux using Win32 and X11: a dice roller, to-do application, number-guessing game, and object-oriented ATM simulator.

HONOURS AND LEADERSHIP

- Second Runner-Up, SEMPEC Junior Hardware Category, NUST CEME - custom vector CPU.
- Runner-Up, STEM 2024 - PKR 20,000 prize.
- Vice President, Dr. Samar Mubarakmand Society for Sciences, GCU Lahore | Sep 2023 - Jan 2025.
- Vice President, Dr. Khorana Society for Chemistry, GCU Lahore | 2023 - 2025.

SELECTED CREDENTIALS

- Machine Learning Specialization - DeepLearning.AI and Stanford Online | Dec 2025.
- Introduction to Logic and Critical Thinking Specialization - Duke University | Jul 2026.
- Game Theory - Stanford University and University of British Columbia | Sep 2024.

TECHNICAL SKILLS

Programming: Python, C, C++17, Java, Verilog, MIPS assembly, JavaScript, TypeScript.

Research and data: reproducible pipelines, provenance metadata, checksums, dataset design, computational stylometry, NumPy, Pandas, PyTorch, Jupyter, LaTeX.

Engineering: Proteus, KiCad, EasyEDA Pro, LTspice, Xilinx ISE, Arduino, digital logic, circuit simulation.

Systems and software: Git, Docker, FastAPI, Node.js, React, SQLite, MongoDB, Linux.